

Sarvesh Prasanth Venkatesan

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EDUCATION

University of Southern California

M.S. in Electrical and Computer Engineering - Machine Learning

Los Angeles, CA

2025-2027

Vellore Institute of Technology

B.Tech. in Electronics and Communication Engineering

Vellore, India

2019 - 2023

EXPERIENCE

Dynamic Robotics and Control Laboratory | University of Southern California

Graduate Researcher | Advisor: Prof. Quan Nguyen

Dec 2025 - Present

Los Angeles, CA

- Designing a **demonstration-guided RL** framework for **dexterous manipulation** with a SEA module, formulating task objectives and reward shaping from VR teleoperated demonstrations.
- Developing a **vision-conditioned RL policy** for humanoid loco-manipulation using **Theia** visual backbone, training a **flow matching** motion generator with residual actions for a phaseless **sim-to-real** pipeline in Isaac Lab.
- Deployed loco-manipulation policy using ONNX runtime on Unitree G1, reducing inference latency by ~40%.

Stochastic Robotics Lab | Indian Institute of Science

Project Associate | Advisor: Prof. Shishir Kolathaya

Jul 2024 – Aug 2025

Bengaluru, India

- Designed a **recurrent-transformer**-based generalist quadruped policy leveraging offline imitation learning to fuse diverse locomotion skills, achieving zero-shot skill transfer across 5+ unseen quadruped morphologies. [Webpage]
- Implemented **PPO based RL locomotion** policies for the 70 kg Stoch5 quadruped; reduced **sim-to-real** gap via novel Ground-Reaction Force and adaptive Raibert-inspired rewards, improving push recovery by **75–98%** on 30° low-traction inclines. [Videos]
- Implemented **visual RL** policies for terrain-adaptive locomotion on challenging terrains. [Code]

Mobile Robotics Lab | Indian Institute of Science

Project Associate | Advisors: Prof. Debashish Ghose

Sep 2023 – Jun 2024

Bengaluru, India

- Developed a centralized **UAV-UGV multi-agent system** for large-scale outdoor inspection, with the UAV executing coverage path planning and UGV for autonomous ground level verification. [Video]
- Built onboard autonomy stacks on ROS2 using **cuVSLAM** for VIO, LiDAR based obstacle avoidance, and multi-agent mission coordination in GPS-denied unstructured environments.
- Collaborated on **LocalizeSORT**, a multi-object tracking framework with world-coordinate association, achieving **+15% HOTA** and **+30% prediction accuracy**. [Paper]

Centre for Non-Destructive Evaluation | Indian Institute of Technology Madras

Intern | Advisors: Prof. Prabhu Rajagopal

Sep 2022 – Jul 2023

Chennai, India

- Developed a full software stack for a Shopping Assistance Robot integrating Cartographer SLAM and ROS2 Nav2 with LiDAR for autonomous retail navigation. [Video]
- Implemented MoveIt2 with OMPL for motion planning and YOLO with RGBD depth projection for 3D object pose estimation, enabling end-to-end autonomous grocery pick and place.

PUBLICATIONS

- Narayanan PP, **Sarvesh Prasanth Venkatesan**, Srinivas Kantha Reddy, Shishir Kolathaya. **GRoQ-LoCO: Generalist and Robot-agnostic Quadruped Locomotion Control using Offline Datasets**. *arXiv*.
- Srihari Vemuru, Kumar Ankit, **Sarvesh Prasanth Venkatesan**, Debashish Ghose, Shishir Kolathaya. **Yield Prediction and Counting Using World Coordinates**. In *ASME IMECE-India 2026 Proceedings*. ASME.

PROJECTS

Manipulation VLA ([Github](#))

- Trained a **vision-language-action (VLA)** using frozen LLaVA backbone with DiT based action head for contact-rich manipulation in simulation.
- Benchmarked diffusion policy against flow matching on LIBERO, with flow matching achieving higher task success rate and 2.5x inference speedup across contact-rich manipulation tasks.

Visual Humanoid Locomotion ([Github](#))

- Developed a scandot based **visual RL humanoid locomotion** policy for terrain adaptive walking, based on this [paper](#). [Video]
- Implemented with CNN based depth encoders as an alternative to scandots for terrain perception.
- Fine-tuning GR00T N1.5 on Unitree G1 for language-guided pick-and-place in IsaacSim via behavior cloning.

SKILLS

Programming: Python, C++.

ML/Robotics: PyTorch, JAX, ROS, MoveIt, IsaacSim, IsaacLab, MuJoCo, mjlal, OpenCV, PX4, ONNX.

Software: Docker, Git, AWS, WandB, Fusion360, LaTeX, Ubuntu Linux.